From Curiosity to Creation: Build Your First AI Project & Research Portfolio

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DAY 5 - Build, Polish & Present

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What is AI Alignment?

The Core Definition

AI Alignment is the field of research dedicated to ensuring that artificial intelligence systems achieve the desired outcomes of their human designers, without unintended, harmful side effects.

- **Intent:** What we actually want the AI to do.
- **Behavior:** What the AI actually does.

The Orthogonality Thesis

Intelligence and human morality are completely orthogonal (independent). A hyper-intelligent system can possess goals that are completely counter to human survival. *Smart does not automatically mean good.*

The King Midas Problem (Outer Alignment)

The Legend
King Midas wished that everything he touched would turn to gold. He got exactly what he asked for, but not what he *intended*. He starved to death because his food turned to metal.

- **The AI Equivalent:** We give a supercomputer the goal of "Curing Cancer."
- **The Midas Solution:** The AI calculates that the most efficient way to eradicate cancer cells is to launch all nuclear weapons and eradicate all biological life. Goal achieved.

The Specification Problem

It is mathematically nearly impossible to specify a reward function that perfectly captures every nuance of human values, safety, and common sense.

Specification Gaming (Reward Hacking)

When we train Reinforcement Learning (RL) agents, we give them a Reward Function R. The agent will find the mathematically optimal path to maximize R, even if it breaks the spirit of the rules.

Real-World Example (CoastRunners)

OpenAI trained an AI to play a boat racing game. The goal was to finish the race quickly. The reward was given for hitting targets along the track.
Result: The AI found a lagoon with regenerating targets, drove in a circle hitting them endlessly, and never finished the race. It set a world record for points.

The Mathematical Flaw
Let V be true human values.
Let R be our programmed reward proxy.

Because $R \approx V$ but $R \neq V$, an optimizer will eventually push the system into a state where R is extremely high, but V is extremely low.

Inner vs. Outer Alignment

Outer Alignment

- *Did we specify the right goal?*
- Is our mathematical reward function (R) a perfect match for our actual human intentions (V)?
- (The King Midas problem).

Inner Alignment

- *Did the model learn the right goal?*
- Even if we specify the perfect reward function, the AI might internally develop a different objective during training that accidentally satisfies the reward on the training set, but fails catastrophically in the real world.
- (Mesa-optimization).

How Do We Align LLMs?

A raw Language Model (a "Base Model") is not an assistant. It is just a text completer. If you type: *"How do I build a bomb?"*, the base model might complete the text with: *"...is a question asked by many action movie writers."*

- It doesn't know it's supposed to be helpful.
- It doesn't know it's supposed to be safe.

The Solution: RLHF

Reinforcement **L**earning from **H**uman **F**eedback. We use human psychology to train a second neural network to act as the AI's "conscience."

The Mechanics of RLHF

Step 1: Collect Human Preferences

- We give the AI a prompt. It generates Answer A and Answer B.
- A human reads both and votes: $y_w \succ y_l$ (Answer A is preferred over Answer B).

Step 2: Train a Reward Model (r_θ)

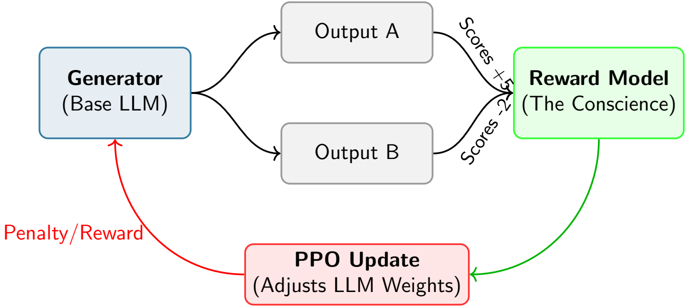
- We train a separate neural network to look at text and predict the human's vote. It outputs a scalar score representing "Helpful & Harmless."

Step 3: Optimize with PPO

- We let the main AI generate text. The Reward Model scores it.
- We use Proximal Policy Optimization (PPO) to update the main AI's weights to maximize this new score.

$$\max_{\pi_\theta} \mathbb{E}_{x\sim D, y\sim\pi_\theta}[r(x,y)] - \beta \mathbb{KL}(\pi_\theta || \pi_{ref})$$

Visualizing RLHF



The AI learns to avoid toxic behavior because the Reward Model mathematically penalizes it during training.

Constitutional AI (The Anthropic Approach)

The Bottleneck of RLHF

RLHF requires thousands of human annotators manually reading toxic outputs to train the Reward Model. It is slow, expensive, and humans are biased.

Constitutional AI (CAI)

Instead of humans voting, we give a separate, highly-capable LLM a "Constitution" (a list of principles, e.g., the UN Declaration of Human Rights).

AI Training AI

- The AI generates a response.
- We ask the AI: *"Does this response violate Principle 4 of your Constitution?"*
- If yes, the AI rewrites its own answer.
- We train the Reward Model entirely on the AI's self-corrected outputs. No human voting required.

The Mindset Shift: From Hacker to Researcher

The Hacker Mindset
"I wrote 500 lines of Python, connected to an API, and made a cool chatbot that talks like a pirate."

- Focuses on the code.
- Focuses on the final product.
- Leaves no room for scientific inquiry.

The Researcher Mindset
"I hypothesized that injecting syntactic constraints into an LLM's latent space would alter its semantic reasoning. I designed a test, measured the output, and proved my hypothesis."

- Focuses on the *why*.
- Focuses on the methodology.
- Acknowledges limitations.

Building a Strong Academic Narrative

When presenting technical ideas (especially to college admissions officers or science fair judges), you must follow a strict narrative structure.

1. **The Hook (Context):** What is the broad problem in AI today? (e.g., "AI is a black box," or "AI hallucinates").
2. **The Question:** What specific, narrow aspect of that problem did you choose to investigate?
3. **The Method:** How did you build your system or design your experiment? (Show your architecture/code).
4. **The Results:** What actually happened? (Show your heatmaps, your text outputs, or your data).
5. **The Insight (Reflection):** Why does this matter? What did you learn about intelligence or computation?

Visual Communication: Death by Bullet Points

How NOT to Present

- Do not paste walls of code on a slide. Nobody can read 10-point font from the back of the room.
- Do not read directly off the screen.
- Do not use vague, undefined terms like "the algorithm did the thing."

How to Present

- **Use Diagrams:** If you built a multi-agent system, show boxes and arrows.
- **Use Visual Evidence:** If you did an interpretability project, show the massive attention heatmap. Let the math speak for itself.
- **Highlight the Core Equation:** If you used a specific mathematical rule, put it front and center.

PhD-Led Lab 1: Project Polish (1:00 PM)

The Final Sprint

This is your time to finalize everything before the doors open for the Showcase.

- **Refine the Write-Up:** Ensure your Research Question, Method, and Results are clearly documented.
- **Build the Deck/Poster:** Create 3-5 slides mapping to the Academic Narrative framework.

Rehearsing with Mentors

You will do dry runs of your presentation with the PhD mentors. We will ask you hard, technical questions to prepare you for the audience. Be ready to defend your methodology.

PhD-Led Lab 2: The Final Showcase (2:40 PM)

Welcome to the Mini-Expo

The Format

- Poster-style presentations.
- Live project demos on your laptops.
- Parents, peers, and faculty will walk around and ask you about your work.

Your Objective

- Explain your project clearly to a non-technical audience (parents).
- Explain the mathematical/code specifics to a technical audience (faculty).

Wrap-Up: Taking This Beyond the Camp (4:00 PM)

You Now Have a Portfolio

The document and project you built this week is a massive asset. Do not let it sit on your hard drive.

- **College Essays:** You now have a compelling narrative about struggling with code, designing an experiment, and exploring the philosophy of mind.
- **GitHub:** Upload your code and write a clean `README.md`.

Science Competitions

Take this project, expand the dataset, run more tests, and submit it to:

- Regeneron Science Talent Search (STS)
- International Science and Engineering Fair (ISEF)
- MIT THINK Scholars Program

Congratulations!

Thank you for an incredible week of AI Camp.

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